Set Theory

A well defined collection of distinct objects is defined to be a set when the objects of collection are independent of order of their arrangement. By a well-defined collection we mean that, their exists a definite rool with the help of which it is possible to tell wheather a particular object belongs to the collection or not. The objects constituting the set are called elements or members of the set.

We shall denote the sets by capital letter A, B, C, X, Y, Z, P etc. and their elements by small letter a, b, c, x, y, z, p, q etc.

If an object is a member of set A, then we write $a \in A$ which may be read as 'a belongs to A' or 'a is an element of A'. On the other hand, if a is not a member of set A, then we write $a \notin A$.

$N = \{\text{Set of all natural numbers}\} = \{1, 2, 3, 4, 5, 6 \dots\}$

$Z^+ = \{\text{Set of all +ve integers}\} = \{1, 2, 3, 4, 5, 6 \dots\}$

$Z^- = \{\text{Set of all -ve integers}\} = \{-1, -2, -3, -4 \dots\}$

$Z = \{\text{Set of all integers}\} = Z^+ \cup Z^- \cup \{0\}$

$Q = \{\text{Set of all rational numbers}\} =$

(i) p/q where p & q are integers.

(ii) $q \neq 0$

(iii) p, q should be co-prime.

$Q^c = \{\text{Set of all irrational numbers}\} = \{\sqrt{2}, \sqrt{3}, e \dots\}$

R = {Set of all real numbers} = $Q \cup Q^c$

$\mathbb{C}$ = {Set of all complex numbers}

: / | → Such that

$\forall$ → for all

$\in$ → Belongs to.

Representation of a set:

(i) Set builder form:

(ii) Tabular form/ Roster form:

A set of all vowels — $A = \{i, e, a, o, u\}$

$B = \{x \in N : 2 < x < 5\}$ — $B = \{3, 4\}$

$C = \{x \in N : 2 < x \leq 7\}$ — $C = \{3, 4, 5, 6, 7\}$

Cardinality: Number of elements present in the set.

$A = \{a, e, i, o, u\}$

$n(A) = 5$

Null Set: A set, having no elements in it is called null set or void set or empty set. It is usually denoted by ϕ.

$A = \phi$ ✓ $A = \{\phi\}$ ✗

Singleton Set: A set consisting of a single element is called a singleton.

Finite set and Infinite set : A set having finite no. of elements is called Finite set. Else it is called infinite set.

Subset : If A and B are two sets such that every element of A is also an element of B, then A is said to be a subset of B. Symbollically this relation is denoted by writing $A \subseteq B$ and is read as 'A is a subset of B' or 'A contained in B'.

Improper Subset : From the definition of subset it is obvious that every set is a subset of itself. i.e. $A \subseteq A$. Here A is called the improper subset of A.

~~B, then we denote this by $A \subset B$.~~

Any other subset of A is said to be a proper subset of A.

Proper Subset : When A is a proper subset of B, then we denote this by $A \subset B$.

Theorem 1 : If $A \subseteq B$ and $B \subseteq A$ then $A = B$

If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.

Power set : If S is any set, then the set of all subsets of S is called the power set of S and is denoted by $P(S)$. Obviously ϕ, S are both the elements of $P(S)$.

$S = \{1, 2, 3\}$

$P(S) = \{\{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\}, \phi\}$.

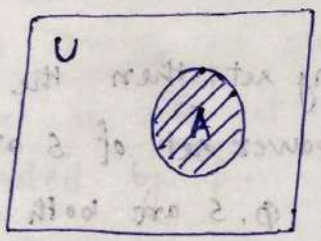
- Cardinality of power set $= 2^n$

  n = no. of elements.

Universal Set: A non-empty set of which all these sets under consideration are subjects is called the universal set, and is denoted by U.

Venn - Euler Diagram:

John - venn, an english logician (1834 - 1923) invented the diagram to present pictorial representation of sets and the operations on set also. These pictures consists of rectangle and closed curves. These combinations of rectangles and closed regions are called venn - Eular diagram. In venn - diagram we shall denote the universal set U by a large rectangle and we write the 'U' in one corner of the rectangle. Also we denote any subset A of the universal set by closed circle or curve and write the letter A within the circle.

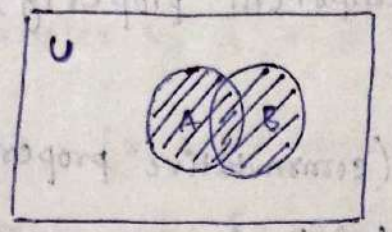


Set operations

1. **Union of sets**: The union of two sets A and B is the set of all elements which are in the set A or in B or in both. This set is denoted by $A \cup B$. Thus

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

The union of two sets A and B is shown with the shaded area in the adjoining Venn diagram.



Using the Venn-diagram the following properties can be easily verified.

(i) $A \cup A = A$ (idempotent property)

(ii) $A \cup \phi = A$

(iii) $A \cup B = B \cup A$ (commutative property).

(iv) If $A \subseteq B$ then $A \cup B = B$.

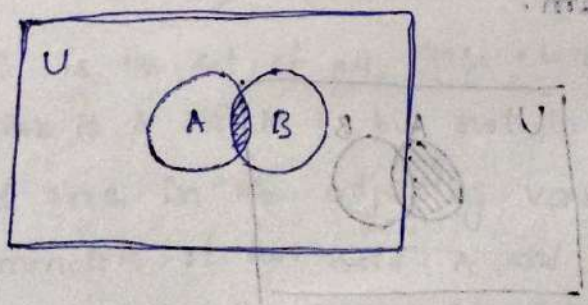
(v) $A \subseteq B \cup A$, $B \subseteq A \cup B$.

(vi) $(A \cup B) \cup C = A \cup (B \cup C)$ (Associative property).

2. Intersection of sets : The intersection of two sets A and B is the set of all those elements which are both in A and B and is denoted by $A \cap B$. Thus

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

The intersection of two sets A and B is shown with shaded area in the adjoining venn diagram.



Using the Venn diagram the following properties can be easily verified.

(i) $A \cap A = A$ (idempotent property).

(ii) $A \cap \phi = \phi$

(iii) $A \cap B = B \cap A$ (commutative property).

(iv) If $A \subseteq B$, then $A \cap B = A$

(v) $A \cap B \subseteq A$, $A \cap B \subseteq B$

(vi) $(A \cap B) \cap C = A \cap (B \cap C)$ (Associative property).

Disjoint sets: Two sets A and B are said to be disjoint if they have no common element i.e. $A \cap B = \phi$.

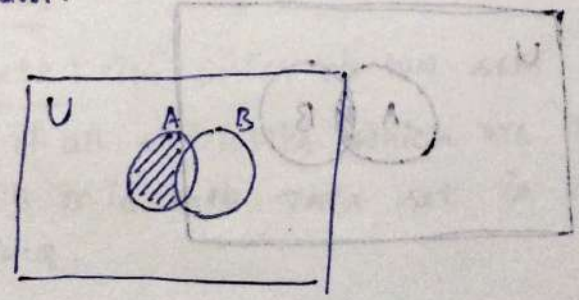
3. Difference of two sets:

The difference of two sets A and B in that order is the set of all those elements of A which do not belong to B, and is denoted by $A - B$.

Thus,

$$A - B = \{x : x \in A \text{ and } x \notin B\}$$

$A - B$ is also called the complement of B with respect to A. The difference $A - B$ of two sets A and B is shown with shaded area in the adjoining Venn diagram.

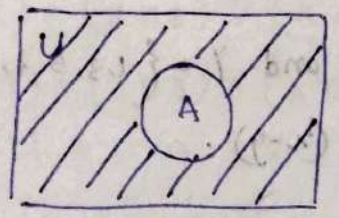


4. **complement of a set relative to the universal set :** The complement of a set A with respect to the universal set U is the difference of U and A and is denoted by $\bar{A}$ or A' or A^c. Thus

$$\bar{A} = U - A = \{x : x \in U \text{ and } x \notin A\}$$

or simply, $\bar{A} = \{x : x \notin A\}$

The complement of a set A is shown with shaded area in the adjoining venn diagram.



Properties :

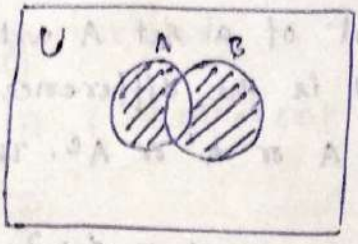
1. $\bar{U} = \phi$.
2. $\bar{\phi} = U$.
3. $A \cap \bar{A} = \phi$.
4. $A \cup \bar{A} = U$.
5. $\bar{\bar{A}} = A$.

5. **Symmetric difference of two sets :** The symmetric difference of two set A and B and is defined by

$$A \Delta B = (A - B) \cup (B - A)$$

Thus $A \Delta B$ is the set of all those elements which belongs either to A or to B but not to both.

The shaded area in the adjoining venn diagram is the symmetric of two sets A and B.



Properties:

1. $A \Delta \phi = A$, $A \Delta A = \phi$.
2. $A \Delta B = B \Delta A$
3. $A \Delta B = (A \cup B) - (A \cap B)$.

If $X = \{1, 2, 3, 4, 5, 6\}$ and $Y = \{1, 3, 5, 7, 9\}$, then find out $(x-y) \cap (y-x) \cup (x-y)$...

$X = \{1, 2, 3, 4, 5, 6\}$

$Y = \{1, 3, 5, 7, 9\}$

$X - Y = \{2, 4, 6\}$

$Y - X = \{7, 9\}$

$(X-Y) \cap (Y-X) = \phi$

$\phi \cup (X-Y) = (X-Y)$.

$= \{4, 2, 6\}$.

If $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $B = \{2, 3, 4, 5\}$, $C = \{2, 4, 6, 8\}$ $D = \{4, 6, 6, 7\}$, find $(B \cup C) \cup (A \cup D)$.

$(A \cup D) = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

~~$(A \cup D) =$~~

$(B \cup C) = \{2, 3, 4, 5, 6, 8\}$

$(B \cup C) \cup (A \cup D) = A$

$= \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

If $A = \{1, 2, 3, 4, 5\}$, $B = \{1, 2, 4, 7, 8\}$ and $C = \{1, 2, 5, 7, 9\}$

Find $(A \cup B) - C$.

$A \cup B = \{1, 2, 3, 4, 5, 7, 8\}$

$(A \cup B) - C = \{4, 8\}$.

Laws of the Algebra of Sets

1. $A \cap (A \cup B) = A$. $A \cup (A \cap B) = A$ (absorptive property).

2. (a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

   (b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (distributive property).

Proof. (a).

First target is to show $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$

Let $x \in A \cap (B \cup C)$

$\Rightarrow x \in A$ and $x \in (B \cup C)$

$\Rightarrow x \in A$ and ($x \in B$ or $x \in C$ or both)

$\Rightarrow (x \in A)$ and $(x \in B)$ or $(x \in A)$ and $(x \in C)$ or both.

$\Rightarrow x \in A \cap B$ or $x \in A \cap C$

$\Rightarrow x \in (A \cap B) \cup (A \cap C)$.

$\therefore$ Therefore, $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ — (1)

Now we need to show. $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$

Let $y \in (A \cap B) \cup (A \cap C)$

$\Rightarrow y \in A \cap B$ or $y \in A \cap C$

$\Rightarrow (y \in A)$ and $(y \in B)$ or $(y \in A)$ and $(y \in C)$ or both.

$\Rightarrow y \in A$ and ($y \in B$ or $y \in C$ or both)

$\Rightarrow y \in A \cap (B \cup C)$

So. $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$ — (ii)

From (i) and (ii). we can prove that

$$\boxed{A \cap (B \cup C) = (A \cap B) \cup (A \cap C)}$$

3. (a) $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$ (De-Morgan's Law).

(b) $\overline{(A \cap B)} = \bar{A} \cup \bar{B}$

Proof: (b) Let $x \in \overline{A \cap B}$

then. $x \in \overline{A \cap B} \iff x \notin (A \cap B)$

$\iff x \notin A$ or $x \notin B$

$\iff x \in \bar{A}$ or $x \in \bar{B}$

$\iff x \in \bar{A} \cup \bar{B}$.

Hence, $\overline{A \cap B} = \bar{A} \cup \bar{B}$.

4. (a) $A - (B \cup C) = (A - B) \cap (A - C)$ (generalization of De-Morgan's Law).

(b) $A - (B \cap C) = (A - B) \cup (A - C)$

Proof. (a) Let $x \in A - (B \cup C)$

$\iff x \in A$ and $x \notin (B \cup C)$

$\iff x \in A$ and $(x \notin B$ and $x \notin C)$

$\iff x \in A$ and $(x \notin B)$ and $(x \in A)$ and $(x \notin C)$

$\iff x \in (A - B)$ and $x \in (A - C)$

$\iff x \in (A - B) \cap (A - C)$.

Hence $A-(B \cup C) = (A-B) \cap (A-C)$

(Proved).

Laws

1. **Idempotent Law:**
   (a) $A \cup A = A$
   (b) $A \cap A = A$

2. **Associative Law:**
   (a) $(A \cup B) \cup C = A \cup (B \cup C)$
   (b) $(A \cap B) \cap C = A \cap (B \cap C)$

3. **Commutative Laws:**
   (a) $A \cup B = B \cup A$
   (b) $A \cap B = B \cap A$

4. **Distributive Laws:**
   (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
   (b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

5. **Identity Laws:**
   (a) $A \cup \phi = A$
   (b) $A \cup U = U$
   (c) $A \cap U = A$
   (d) $A \cap \phi = \phi$.

6. **Involution Laws:**
   $(A^c)^c = A$.

7. **Complement Laws:**
   (a) $A \cup \bar{A} = U$
   (b) $\bar{U} = \phi$
   (c) $A \cap \bar{A} = \phi$
   (d) $\bar{\phi} = U$.

8. De Morgan's Laws :

(a) $\overline{A \cup B} = \bar{A} \cap \bar{B}$

(b) $\overline{A \cap B} = \bar{A} \cup \bar{B}$

For any three sets A, B, C show that $A - (B - C) = (A - B) \cup (A \cap C)$.

Let, $x \in A - (B - C)$

$\Leftrightarrow x \in A$ and $x \notin (B - C)$

$\Leftrightarrow x \in A$ and $(x \notin B$ or $x \in C)$

$\Leftrightarrow (x \in A)$ and $(x \notin B)$ or $(x \in A)$ and $(x \in C)$

$\Leftrightarrow x \in (A - B)$ or $x \in (A \cap C)$

$\Leftrightarrow x \in (A - B) \cup (A \cap C)$.

If A, B are non-empty sets, then prove that $(A - B)$ and $(A \cap B)$ are pairwise disjoint sets.

Let, $x \in (A - B) \cap (A \cap B)$

$\Rightarrow x \in (A - B)$ and $x \in (A \cap B)$

$\Rightarrow ((x \in A)$ and $(x \notin B))$ and $((x \in A)$ and $(x \in B))$

But there is no element x which satisfied both $x \in B$ and $x \notin B$. Hence there is no element in $(A - B) \cap (A \cap B)$

So, $(A - B) \cap (A \cap B) = \emptyset$.

- **Ordered Pair:** Let A, B be two non-empty sets and a belongs to A, b belongs to B. Then (a, b) is called an ordered pair and a, b are called the first and second co-ordinate of the ordered pair (a, b). If (a, b) and (c, d) are two ordered pairs; then (a, b) = (c, d) if and only if ~~a = b~~ a = c and b = d.

- **Cartesian Product of two sets:** Let A and B be two non-empty sets. Their cartesian product denote by A × B is the set of all distinct ordered pairs where first co-ordinate is an element of A and the second co-ordinate is an element of B. Thus

$$A \times B = \{(a, b) : a \in A \& b \in B\}$$

* In ordered pair $(a, b) \neq (b, a)$.

If A and B are non-empty sets; then cardinality of A and B are 2 & 3 respectively. Then cardinality of A × B is - 6.

If A = {1, 2} and B = {2, 3}, C = {3, 4} then find

(i) $(A \times B) \cup (A \times C)$

(ii) $(A \times B) - (A \times C)$.

$A \times B = \{\{1,2\}, \{1,3\}, \{2,2\}, \{2,3\}\}$

$A \times C = \{\{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}\}$

(i) $(A \times B) \cup (A \times C) = \{(1,2), (1,3), (1,4), (2,2), (2,3), (2,4)\}$

(ii) $(A \times B) - (A \times C) = \{(1,2), (2,2)\}$

If $A = \{1, 2, 3\}$, $B = \{4, 5\}$, $C = \{1, 2, 3, 4, 5\}$, then $(C \times B) - (A \times B)$ –

$C \times B = \{(1,4), (1,5), (2,4), (2,5), (3,4), (3,5), (4,4), (4,5), (5,4), (5,5)\}$

$A \times B = \{(1,4), (1,5), (2,4), (2,5), (3,4), (3,5)\}$

$(C \times B) - (A \times B)$

$= \{(4,4), (4,5), (5,4), (5,5)\}$

Prove that, $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$.

Let (x, y) be any arbitrary element of $(A \times B) \cap (C \times D)$

Then, $(x, y) \in (A \times B) \cap (C \times D)$

$\Leftrightarrow (x, y) \in (A \times B)$ and $(x, y) \in (C, D)$

$\Leftrightarrow (x \in A$ and $y \in B)$ and $(x \in C$ and $y \in D)$

$\Leftrightarrow (x \in A$ and $x \in C)$ and $(y \in B$ and $y \in D)$

$\Leftrightarrow x \in (A \cap C)$ and $y \in (B \cap D)$

$\Leftrightarrow (x, y) \in (A \cap C) \times (B \cap D)$.

$\therefore (A \times B) \cap (C \times D) = (A \cap C) \times (B \times D)$.

1 Marks

1. If A & B are two disjoint sets then
   (i) $\{A \cap B\} = \{A\}$
   (ii) $\{A \cap B\} = \{B\}$ $\Rightarrow \phi$.
   (iii) $\{A \cap B\} = \{\phi\}$
   (iv) $A \cap B = \{U\}$

2. If $A \subseteq B$ then $A \cup B =$ B.

3. If A and B are any two sets then $A \cap (\overline{A \cap B}) =$ A − B.

4. The no. of subsets of a set with n elements is 2^n.

5. Null set is a subset of every set.

6. Cardinality of power set of a non-empty set A is - $2^{|A|}$.

7. which is a singleton $\{0\}$.

8. If A is a proper subset of a non-empty set S and two subsets A and $\bar{A}$ are non-empty. Then which one is true?

   $A \cup \bar{A} = S$ & $A \cap \bar{A} = \phi$.

Relation

Let A and B be two non-empty sets. A relation between two sets A and B is a subset of $A \times B$ and is denoted by R (or ρ). Thus $R \subseteq A \times B$. When $R \subseteq A \times B$, then R is said to be a relation on the set A.

If $(a, b) \in R$ then obviously $(a, b) \in A \times B$ and we write aRb and read as 'a is related to b'.

Again if $(a, b) \notin R$ then we write $a (\sim R) b$ or $a \bar{R} b$ and read as 'a is not related to b'. We may describe a relation R in the set builder notation as -

$$R = \{(a, b) : a \in A, b \in B\}.$$

Example

$A = \{2, 4, 6\}$ & $B = \{4, 6, 18\}$, $R = \{(a, b) : a \in A, b \in B$ and a is a factor of $b\}$.

$$A \times B = \{(2,4), (2,6), (2,18), (4,4), (4,6), (4,18), (6,4), (6,6), (6,18)\}.$$

check for every pair if it satisfies the condn.

$$R = \{(2,4), (2,6), (2,18), (4,4), (6,6), (6,18)\}$$

Example :

$A = \{1, 2, 3, 4\}$

$A \times A = \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4) \ldots\}$

$R = \{(a, b) \in A \times A \ \& \ a \text{ is the square of } b\}$

$R = \{(1, 1), (4, 2)\}$.

Different types of Relation :

1. **Void Relation :** If $R = \phi$ (void set), then $a R b$ does not hold for any pair of elements a, of a set A. This relation R is said to be void or null relation in A.

2. **Inverse Relation :** Let R be a relation from A to B. Then the inverse of R, denoted by R^{-1}, is a relation from B to A is such that $b R a$ hold if $a R b$ holds. That is

$$R^{-1} = \{(b, a) : b \in B, a \in A \text{ and } (a, b) \in R\}.$$

Ex. $A = \{1, 2\}$ $B = \{2, 3\}$

$R = \{(1, 2), (2, 3)\}$

then $R^{-1} = \{(2, 1), (3, 2)\}$.

3. **Reflexive Relation :** Let R be a relation from A. Then R is said to be a reflexive relation if:-

$(a, a) \in R$

i.e. if $a R a \ \forall \ a \in A$

4. **Symmetric Relation :** Let R be a relation from A. Then R is said to be a symmetric relation if:-

$(a, b) \in R \Rightarrow (b, a) \in R \ \forall \ (a, b) \in R$

i.e. if $a R b \Rightarrow b R a$.

5. Transitive Relation: Let R be a relation from A. Then R is said to be a transitive relation if.

$(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R$

i.e. if aRb & $bRc \Rightarrow aRc$.

Equivalence Relation: Let R be a relation on a set A. Then R is said to be an equivalence relation if R is reflexive, symmetric and transitive.

Number of relations from A to A:

$= 2^{\text{Number of elements } A \times A}$

$= 2^{\text{No. of elements in A} \times \text{No. of elements in A}}$

$= 2^{n^2}$

Number of reflexive relations from A to A: 2^{n^2-n}

Number of symmetric relations from A to A:

$2^{\frac{n(n+1)}{2}}$

■ Domain and Range of a Relation: The domain of a relation is the set of all first-coordinate of the ordered pairs in the relation, and the range is the set of all second coordinate of the ordered pairs in the relation.

Ex. If $A = \{2, 3\}$, $B = \{1, 2, 3, 4\}$ then $R = \{(a, b) : a \in A, b \in B \text{ and } a \leq b\}$.

$R = \{(2, 2), (2, 3), (2, 4), (3, 3), (3, 4)\}$

Domain of $R = \{2, 3\}$

Range of $R = \{2, 3, 4\}$

Ex. If R is the relation from $A = \{1, 2, 3, 4\}$ to $B = \{2, 3, 4, 5\}$. Find the elements in R defined by $a\,R\,b$, if a and b are both odd. Write also the domain and range of R.

$R = \{(1,3), (1,5), (3,3), (3,5)\}$

Domain $= \{1, 3\}$

Range $= \{3, 5\}$

Determine the nature of the following relations on the set $A = \{1, 2, 3\}$

(i) $R = \{(3,3)\}$ (ii) $R = \{(1,2), (2,2)\}$

(iii) $R = \{(1,1), (1,2), (2,3), (2,2), (3,3), (2,1), (3,2)\}$

(i) Not reflexive, symmetric, transitive.

(ii) Not reflexive, not symmetric, transitive.

(iii) Reflexive, Symmetric, Not transitive.

Show that the following relation on Z is symmetric, transitive but not reflexive: $R = \{(a,b) : a, b$ is integer and $a.b > 0\}$.

Let $a, b \in \mathbb{Z}$ such that $a\,R\,b$ holds.

i.e. $(a,b) \in R \quad \therefore a.b > 0$

$\Rightarrow b.a > 0$

$\Rightarrow (b,a) \in R$

$\Rightarrow b\,R\,a$ holds

$\therefore$ R is symmetric

Again Let $a, b, c \in \mathbb{Z}$ such that $a\,R\,b$, $b\,R\,c$ holds.

$\therefore a \cdot b > 0$ and $b \cdot c > 0$

$\Rightarrow a \cdot b \cdot b \cdot c > 0$

$\Rightarrow a \cdot b^2 \cdot c > 0$

$\Rightarrow ac > 0$ (because b^2 will be always > zero).

$\Rightarrow a R c$ holds

$\Rightarrow (a, c) \in R$

$\therefore$ R is transitive.

Let $a \in z$

now $(a\,a) \in R$ then we should have $a \cdot a > 0$; but if $a^2 > 0$ is not true as we know $a^2 \geq 0$.

i.e. if $a = 0$ then $a^2 = 0$.

$\therefore$ R is not reflexive.

On the set z of integers, define a binary relation R defined by $aR\,b = \{(a, b)$ such that $a+b$ is even for all a, b in $z\}$

Show that R is an equivalence relation.

Let $a \in z$ as $a + a = 2a$ be even for all $a \in z$

So. aRa holds $(a\,a) \in R$

$\therefore$ R is reflexive. — (I)

Again let, $(a, b) \in z$ such that aRb holds

$\therefore (a\,b) \in R$

$\therefore a+b$ = even no.

$\Rightarrow b+a$ = even no.

$(b, a) \in R$, bRa holds

$\therefore$ R is symmetric — (II)

Now let $a, b, c \in \mathbb{Z}$ such that $a R b$, $b R c$ holds.

$(a, b) \in R \Rightarrow a+b$: even no.

$(b, c) \in R \Rightarrow b+c$: even no.

$a+2b+c$: Even no.

$a+c$ = Even no. $- 2b$

= Even no.

$\therefore a R c$ holds

$\therefore$ R is transitive. — (iii)

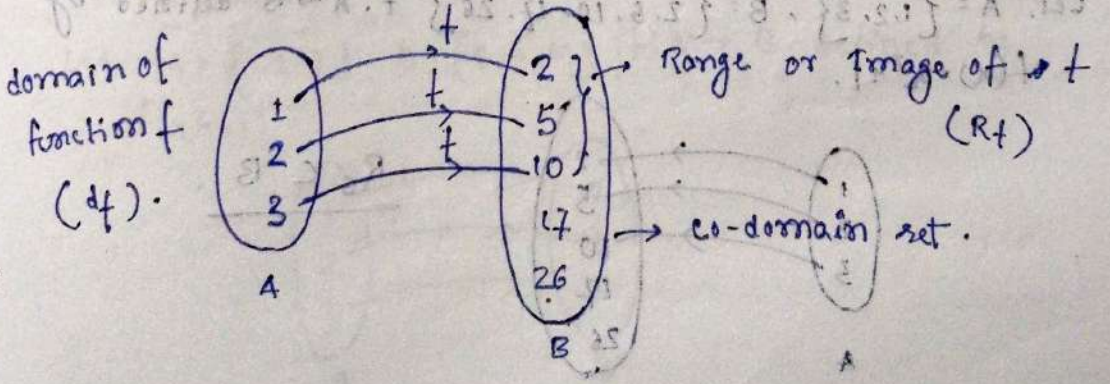
From (i), (ii), (iii) R is equivalence.

Functions

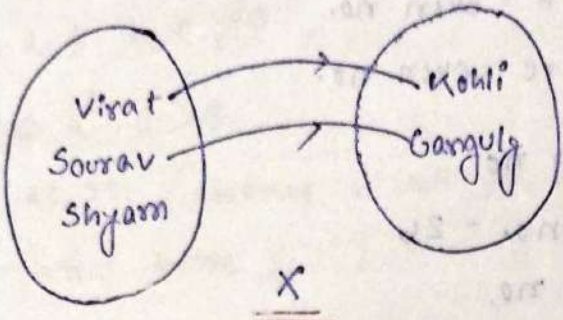
Let A and B be two non-empty sets. If there exists a correspondence i.e. a rule, denoted by f, which associates to each element of A a unique element of B, then f is called a function from A to B and is denoted by $f : A \to B$.

The set A is called the domain of f and B is called the co domain of f. the set of all images under f is called the Range set or Image set and is denoted by $f(A)$. Thus $f(A) = \{f(x) : x \in A\}$ and $f(A) \subseteq B$.

Let $A = \{1, 2, 3\}$, $B = \{2, 5, 10, 17, 26\}$ $f : A \to B$ defined by $f(x) = x^2 + 1$.

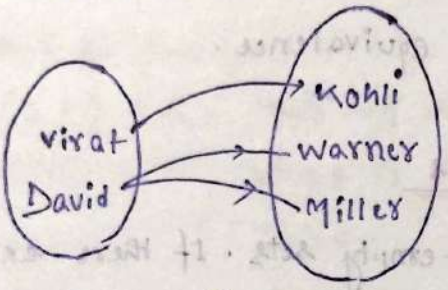


Some cases to remember:

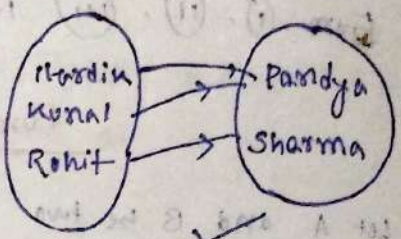


X

Every element of domain set must be connected to one element of co domain set.



X



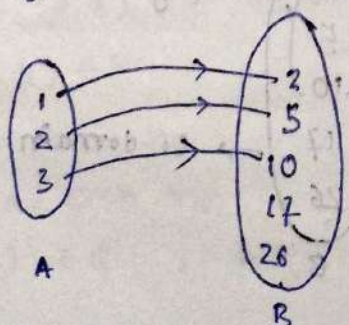
✓

Any element of domain set can't be connected to multiple elements of co-domain set. Vice Versa is true.

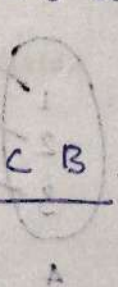
Different types of function

1. **Into function**: A $f: A \to B$ is said to be an into function if the range set $f(A)$ is a proper subset of the co-domain B i.e. if $f(A) \subset B$ and we say that f maps A into B.

Let, $A = \{1, 2, 3\}$, $B = \{2, 5, 10, 17, 26\}$ $f: A \to B$ defined by $f(x) = x^2 + 1$.

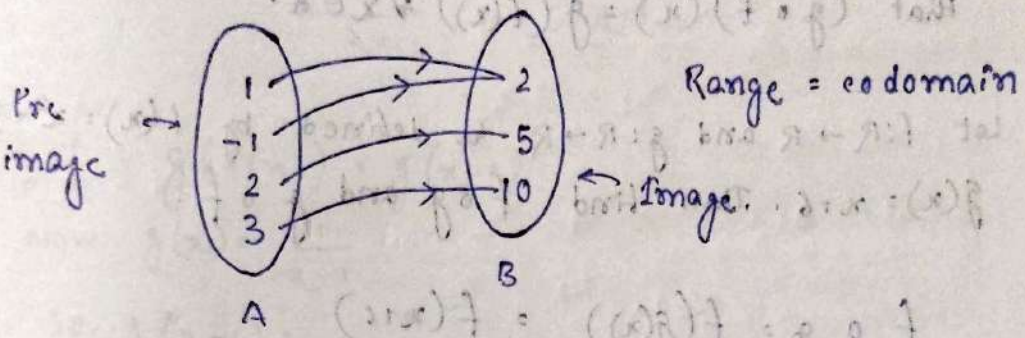


$R_S \subset B$



2. **Onto or Surjective Function :** A function $f: A \to B$ is said to be an onto function if the range set $f(A)$ is equal to the domain B i.e. every element of B is an image of some element of A and we say that f maps A onto B.

Let $A = \{1, -2, 3, 2\}$, $B = \{2, 5, 10\}$ $f: A \to B$ defined by $f(x) = x^2 + 1$.



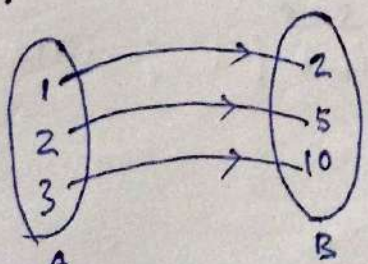
3. **One-to-one or Injective Function :** If $f: A \to B$ is said to be one to one if different elements of A have different images in B i.e. if $x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$

in other words $f(x_1) = f(x_2) \Rightarrow x_1 = x_2 \quad \forall\ x_1, x_2 \in A$.

4. **Many-one Function :** $f: A \to B$ is said to be many-one if ~~different~~ more than one element of A have same image in B.

5. **Bijective Function :** A function $f: A \to B$ is said to be bijective if f is one-to-one and onto.

Let $A = \{1, 2, 3\}$. $B = \{2, 5, 10\}$ $f: A \to B$ defined by $f(x^2 + 1)$.



** If $f: A \to B$ is a bijective function then $n(A) = n(B)$.

6. Composition Function: Let f and g be two functions such that $f: A \to B$ and $g: B \to C$. Then the composition of the function f and g denoted by $(g \circ f)$ is a function given by $(g \circ f): A \to C$, such that $(g \circ f)(x) = g(f(x)) \ \forall x \in A$.

Let $f: R \to R$ and $g: R \to R$ be defined by $f(x) = x^2 + 3$ $g(x) = x + 6$. Then find $f \circ g$ and $g \circ f$.

$$f \circ g = f(g(x)) = f(x+6)$$

Now, $f(x) = x^2 + 3$

So, $f(x+6) = (x+6)^2 + 3$

$= x^2 + 12x + 36 + 3$

$= x^2 + 12x + 39.$

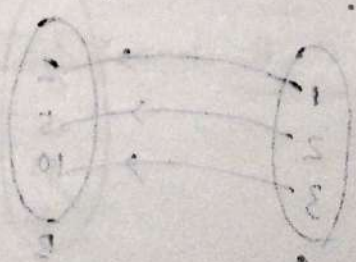
$$g \circ f = g(f(x)) = g(x^2 + 3)$$

Now, $g(x) = x + 6$

So, $g(x^2 + 3) = (x^2 + 3) + 6$

$= x^2 + 9$

** Generally $g \circ f \neq f \circ g$.



If $f(x) = x - 1$ and $g(x) = \frac{1}{x+1}$ then find fog. gof. fof. gog.

$$fog = f(g(x)) = f\left(\frac{1}{x+1}\right)$$

Now. $f(x) = x - 1$

$$f\left(\frac{1}{x+1}\right) = \frac{1}{x+1} - 1$$

$$= \frac{1 - x - 1}{x+1} = -\frac{x}{x+1}$$

$$gof = g(f(x)) = g(x-1)$$

Now. $g(x) = \frac{1}{x-1}$

So. $g(x-1) = \frac{1}{x-1-1} = \frac{1}{x-2}$

$$fof = f(f(x)) = f(x-1)$$

Now. $f(x) = x - 1$

$$f(x-1) = x - 1 - 1$$
$$= x - 2$$

$$gog = g(g(x)) = g\left(\frac{1}{x+1}\right)$$

Now. $g(x) = \frac{1}{x+1}$

So. $g\left(\frac{1}{x+1}\right) = \frac{1}{\frac{1}{x+1} + 1}$

$$= \frac{x+1}{x+2}$$

Show that the following function g is neither subjective nor injective : $g : R \to R$ defined by $g(x) = x^2$, $x \in R$

To prove injective. $f(x_1) = f(x_2) \Rightarrow x_1 = x_2 \ \forall \ x_1, x_2 \in D_f$

Let x_1 and $x_2 \in R$ (domain) be any two arbitrary element such that, $g(x_1) = g(x_2)$

$$\Rightarrow x_1^2 = x_2^2$$

$$\Rightarrow x_1 = \pm x_2$$

$g(x_1) = g(x_2) \Rightarrow x_1 = x_2$

also, $x_1 = -x_2$

$\therefore g$ is not injective.

Let, $y \in R$ (codomain) be any arbitrary element ~~such that~~ $\exists \ x \in R$ (domain) such that $g(x) = y$

$$\Rightarrow x^2 = y$$

$$\Rightarrow x = \pm y.$$

Since, pre-images do not exist for negative real numbers, so g is not surjective or onto.

Let, $A = R - \{3\}$ and $B = R - \{1\}$, R is the set of real numbers. Consider the function defined by $f(x) = \dfrac{x-2}{x-3}$. Is f one to one and onto? justify your answer. Also find f^{-1} if exists.

one to one.

x_1, x_2

$f(x_1) = f(x_2)$

$\Rightarrow \frac{x_1 - 2}{x_1 - 3} = \frac{x_2 - 2}{x_2 - 3}$

$\Rightarrow (x_1 - 2)(x_2 - 3) = (x_1 - 3) \cdot (x_2 - 2)$

$\Rightarrow x_1 = x_2$

$\therefore$ f is injective or one to one.

onto.

$f(x) = y$

$\Rightarrow \frac{x-2}{x-3} = y.$

$\Rightarrow x - 2 = xy - 3y.$

$\Rightarrow x - yx = 2 - 3y$

$\Rightarrow x = \frac{2 - 3y}{1 - y}$

for all $y \in R - \{1\}$ $\exists$ x in $R - \{3\}$.

$\therefore$ f is a onto and surjective function

Since, f is both injective (one one) and surjective (onto). f is a bijective mapping.

So, f^{-1} exists.

Let, $f(x) = y$

$x = f^{-1}(y)$

$\frac{2 - 3y}{1 - y} = f^{-1}(y).$

Find the domain of the function $f(x) = \frac{x^2+1-x}{x^2+5x-6}$

$f(x)$ has real, finite value when

$x^2 + 5x - 6 \neq 0$

$(x-3)(x-2) \neq 0$

$x-3 \neq 0$ $\quad$ $x-2 \neq 0$

$x \neq 3$ $\quad$ $x \neq 2$

domain : $R - \{2, 3\}$.

Find the domain of the function $f(x) = \frac{x^2+1-x}{\sqrt{x^2-5x+6}}$

$x^2 - 5x + 6 > 0$ ✓

$x^2 - 5x + 6 \neq 0$.

Sol^n: The function has real, finite value if

$x^2 - 5x + 6 \geq 0$

Since the denominator should not be $= 0$.

$x^2 - 5x + 6 > 0$

$(x-3)(x-2) > 0$

Now either $(x-3) > 0$ & $(x-2) > 0$

$\therefore x > 3$ or $x > 2$

$\Rightarrow x > 3$

or. $(x-3) < 0$ & $(x-2) < 0$

$\therefore x < 3$ & $x < 2$

$\Rightarrow x < 2$

Domain of this function: $(3, \infty) \cup (-\infty, 2)$

Find the domain of $f(x) = \sqrt{\frac{1-|x|}{2-|x|}}$

The function is real if $\frac{1-|x|}{2-|x|} \geq 0$ — (i)

and if $2-|x| \neq 0$ — (ii) $\Rightarrow |x| \neq 2$

$x \neq \pm 2$ — (v)

(i) $\Rightarrow$ <u>case 1:</u>

$1-|x| \geq 0$ & $2-|x| > 0$

i.e. $|x| \leq 1$ & $|x| < 2$

i.e. $x \leq 1$ $\quad x < 2$

$-x \leq 1$ $\quad x > -2$

$x \geq -1$

$-1 \leq x \leq 1$ & $-2 < x < 2$

$x \in [-1, 1] \cap (-2, 2)$ — (iii)

<u>case 2:</u>

$1-|x| \leq 0$ & $2-|x| < 0$

i.e. $|x| \geq 1$ & $|x| > 2$

i.e. only $|x| > 2$

$x > 2$ or $x < -2$

$x \in (-\infty, -2) \cup (2, \infty)$ — (iv)

from (ii), (iv) & (v),

$[-1, 1] \cup (-\infty, -2) \cup (2, \infty)$

Find the range or image set of the function.

$$f(x) = \frac{1}{2 - \cos 4x}.$$

$$-1 \leq \cos 4x \leq 1$$

$$\Rightarrow 1 \geq -\cos 4x \geq -1$$

$$\Rightarrow 2+1 \geq 2 - \cos 4x \geq -1+2$$

$$\Rightarrow 3 \geq 2 - \cos 4x \geq 1$$

$$\Rightarrow \frac{1}{3} \leq \frac{1}{2 - \cos 4x} \leq 1.$$

$$\boxed{\frac{1}{3} \leq f(x) \leq 1}$$

So, range : $[\frac{1}{3}, 1]$

1 Marks question :

1. Let the function $f: R \to R$ defined by

$$f(x) = 2x - 1 \quad \text{for } x > 2$$
$$= x^2 - 1 \quad \text{for } -2 \leq x \leq 2$$
$$= 3x + 1 \quad \text{for } x < -2.$$

Then the value of $f(-3)$ is

$$= 3.(-3) + 1$$
$$= -8.$$

2. If $f(x+1) = 2x + 3$, the $f(-3)$ is

$$f(-3) = f(-4+1) = 2.(-4) + 3$$
$$= -8 + 3$$
$$= -5$$

3. The function $f(x) = x^2 - 2x + 2$ is

> if $f(-x) = f(x)$ then even function.
> $f(-x) = -f(x)$ then odd function

$$f(-x) = x^2 + 2x + 2$$
$$\neq f(x)$$
$$\neq -f(x)$$

so. it is neither odd function nor even function.

4. If $f(x) = \frac{ax-b}{bx-a}$ then $f(x) \cdot f\left(\frac{1}{x}\right)$ is -

$$f\left(\frac{1}{x}\right) = \frac{a/x - b}{b/x - a}$$
$$= \frac{a - bx}{b - ax}$$

so. $f(x) \cdot f\left(\frac{1}{x}\right) = \frac{ax-b}{bx-a} \times \frac{a-bx}{b-ax}$

$$= + \left(\frac{b-ax}{-bx+a}\right) \times \frac{a-bx}{b-ax}$$
$$= 1$$

5. The function $f(x) = x/(x^2-9)$ can't be defined when

$$x^2 - 9 = 0$$
$$x^2 = 3$$
$$x = \pm 3$$

6. If $f(x) = \frac{|x|}{x}$, then for any $\lambda > 0$,

$|f(\lambda) - f(-\lambda)| = ?$

$$f(x) = \frac{x}{x} = 1 \quad , x > 0$$
$$= -x/x = -1 \quad , x < 0$$

$f(\lambda) = 1 \qquad f(-\lambda) = -1$

So. $|f(\lambda) - f(-\lambda)| = |1+1| = 2$

7. The value of $[x]$ in $3 \leq x < 4$ is - 3.

greatest integer function.

$[x] = 0, \quad 0 \leq x < 1$

$= 1, \quad 1 \leq x < 2$

$= 2, \quad 2 \leq x < 3.$

8. Let $|A| = |B| = n$ where $|A|$ denotes no. of elements of A. then no. of bijective mapping from A to B is - $n!$.

7. If the function $f: R \to R$ defined by $f(x) = 3x - 4, \; x > 0$

$= -3x + 2, \; x \leq 0$

then $f^{-1}(2) = 2$.

12. A relation R is called an equivalence relation if-

R is reflexive, symmetric and transitive.

13. For the mapping $g: [-3, 2] \to R$ defined by $g(x) = 3x + 4$ for any $x \in [-3, 2]$ then image set of g is -

$$-3 \leq x \leq 2$$
$$\Rightarrow -9 \leq 3x \leq 6$$
$$\Rightarrow -9 + 4 \leq 3x + 4 \leq 6 + 4$$
$$\Rightarrow -5 \leq 3x + 4 \leq 10$$

$[-5, 10]$.

14. ρ is a relation defined by on R by "$a \rho b$ if $a \geq b$" where R is real no. set. Then ρ is -

reflexive, anti-symmetric, transitive

15. The mapping $f: N \to N$ defined by $f(n) = \left[\frac{n+1}{2}\right]$, $n \in N$ is - surjective.

16. The mapping $f: R \to R$ defined by $f(x) = (x^2+1)^{2014}$ then the mapping is

not injective, not surjective

Principle of Mathematical Induction

The well ordering property:

The set N consisting of numbers 1, 2, 3 ... is called the set of all natural numbers. The well ordering property of the set N states that every non-empty subset of N contains at least one element. This means that if S be a non-empty subset of N, there is some natural number a in S such that $a \leq x$ for all x in S.

Principle of Induction: Let S be a subset of N with the properties -

1. 1 belongs to S (basis of induction).
2. whenever a natural no. K belongs to S, then K+1 belongs to S (Induction hypothesis).

Then $S = N$

Prove by mathematical induction:

$$1 + 2 + 3 + \dots + n = n(n+1)/2$$

Let $P(n)$ be the statement where,

$$P(n): \quad 1 + 2 + 3 + \dots + n = n(n+1)/2$$

Now, for $n = 1$;

$$1 = 1 \cdot (1+1)/2 = 1$$

$\therefore P(n)$ is true for $n = 1$.

Now, Let us assume that the statement is true for $n = m$.

$$\therefore 1 + 2 + 3 + \dots + m = m(m+1)/2 \quad \text{— (1)}$$

Now, for $n = m+1$

$LH.S = 1+2+3+\ldots+m+m+1$

$= \frac{m(m+1)}{2} + (m+1)$ for [from (1)]

$= (m+1)\left[\frac{m}{2}+1\right]$

$= \frac{(m+1)(m+2)}{2} = RHS.$

$RH.S = \frac{n(n+1)}{2} = \frac{(m+1)(m+2)}{2}$

∴ The statement $p(n)$ is true for $n = m+1$ whenever it is true for $n = 1$ & $n = m$. Hence, $1+2+3+\ldots+n = n(n+1)/2$ $\forall n \in N$.

Use mathematical induction to prove that n^3+2n is divisible by 3.

Let $p(n)$ be the statement where:

$p(n) = n^3+2n$ is divisible by 3 $\forall n \geq 1$

Now, for $n=1$,

$n^3+2n = 1+2+3$, which is divisible by 3.

∴ $p(n)$ is true for $n=1$.

Now, Let us assume that $p(n)$ is true for $n=m$.

$\therefore n^3+2n = m^3+2m$ is divisible by 3.

$= 3k$ where k is any integer.

——— (1)

For, $n = m+1$.

$n^3+2n = (m+1)^3+2(m+1)$

$= m^3+3m^2+3m+1+2m+2$

$= m^3+2m+3m^2+3m+3$

$= 3k+3m^2+3m+3$

$= 3[k+m^2+m+3] = 3k_1$ where k_1 is integer.

6 ∴ The statement is true for $n=m+1$ whenever it is true for $n=1$ and $n=m$. Hence, n^3+2m is divisible by 3, $n \geq 1$.

Prove by mathematical induction, $6^{n+2}+7^{2n+1}$ is divisible by 43 for each positive integer n.

Let, $P(n)$ be the statement where,

$P(n): 6^{n+2}+7^{2n+1}$ is divisible by 43. $\forall n \in N$

Now, for $n=1$.

$= 6^{1+2}+7^{2+1}$

7 $= 6^3+7^3$

$= 216+343$

$= 559$

which is divisible by 43.

$\therefore P(n)$ is true for $n=1$.

Let us assume, $P(n)$ is true for $n=m$.

8 $= 6^{m+2}+7^{2m+1}$ is divisible by 43.

$= 6^{m+2}+7^{2m+1} = 43k$ for some integer k.

——— (1)

7 Now, for $n=m+1$.

$= 6^{m+3}+7^{2m+3}$

$= 6^{m+2} \cdot 6 + 7^{2m+1} \cdot 7^2$

$= 6 \cdot 6^{m+2} + 49 \cdot 7^{2m+1}$

$= 6\left(43k - 7^{2m+1}\right) + 49 \cdot 7^{2m+1}$

$= 6 \cdot 43k + 7^{2m+1} \cdot 43$

$= 43\left[6k + 7^{2m+1}\right] = 43k_1$, where k_1 is int.

- which is divisible by 43.

∴ The statement is true for $n = m+1$ whenever it is true for $n = 1$ and $n = m$.

Hence, $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each positive integer n.

Prove that: $n! > 2^n$ for all natural numbers $n \geq 4$.

Let $P(n)$ be the statement where,

$$P(n) : n! > 2^n \quad \forall \quad n \geq 4$$

for $n = 4$,

$$4! = 4 \cdot 3 \cdot 2 \cdot 1$$
$$= 24 > 2^4$$
$$= 24 > 16$$

∴ $P(n)$ is true for $n = 4$.

Let us assume that, $P(n)$ is true for $n = m$.

$$m! > 2^m \quad \text{---} (1)$$

Now, for $n = m+1$,

$$(m+1)! = (m+1) \cdot m! > (m+1) 2^m$$
$$= \cancel{(m+1) \cdot 2^m}$$
$$> 2^{m+1}$$

∴ $P(n)$ is true for $n = m+1$ whenever it is true for $n = 4$ and $n = m$.

Hence, $n! > 2^n$ is for all natural numbers $n \geq 4$.

6. ■ Second principle of induction: Let S be a subset of N such that

(i) $1 \in S$: and

(ii) If $\{1, 2, \dots k\} \in S$. Then $k+1 \in S$. Then $S = N$.

Prove that for all $n \in N$. $(2+\sqrt{3})^n + (2-\sqrt{3})^n$ is an even integer.

Let $P(n)$ be the statement where

$P(n)$: $(2+\sqrt{3})^n + (2-\sqrt{3})^n$ is an even integer, $\forall\ n \in N$.

7. For, $n = 1$,

$(2+\sqrt{3})^1 + (2-\sqrt{3})^1$

$= 4$. which is an even integer

$\therefore$ $P(n)$ is true for $n = 1$.

Let. $P(n)$ is true for $n = 1, 2, 3, \dots k$.

Now. for $n = k+1$.

8. $(2+\sqrt{3})^{k+1} + (2-\sqrt{3})^{k+1}$

$= a^{k+1} + b^{k+1}$ (where, $a = 2+\sqrt{3}$. $b = 2-\sqrt{3}$)

7. $= (a^k + b^k)(a+b) - (a^{k-1} + b^{k-1}) \cdot ab$

$= 4 \cdot (a^k + b^k) - 1 \cdot (a^{k-1} + b^{k-1})$.

= even integer, since $(a^k + b^k)$ & $(a^{k-1} + b^{k-1})$ is also an even integer.

$\therefore$ $P(k+1)$ is true whenever $P(1), P(2), \dots P(k)$ are true. Then by 2nd principle of mathematical induction. for all $n \in N$. $(2+\sqrt{3})^n + (2-\sqrt{3})^n$ is an even integer.

Prove by mathematical induction.

$$1^2+2^2+3^2+\dots+n^2 = \frac{n(n+1)(2n+1)}{6} \quad \forall n \in \mathbb{N}$$

Let, $P(n)$ be the statement where,

$$P(n): 1^2+2^2+3^2+\dots+n^2 = \frac{n(n+1)(2n+1)}{6}, \forall n \in \mathbb{N}$$

Now, for $n=1$,

$$1^2 = \frac{1.(1+1).(2+1)}{6}$$

$$= \frac{1.2.3}{6} = 1$$

So. $P(n)$ is true for $n=1$.

Let us assume that, $P(n)$ is true for $n=m$.

$$1^2+2^2+\dots+m^2 = \frac{m.(m+1)(2m+1)}{6} \quad \text{---(1)}$$

Now, for $n=m+1$,

LHS: $1^2+2^2+\dots+m^2+(m+1)^2 =$ ~~$(m+1)($~~

$$= \frac{m.(m+1)(2m+1)}{6} + (m+1)^2$$

$$= (m+1)\left[\frac{m(2m+1)}{6} + (m+1)\right]$$

$$= (m+1).\left(\frac{2m^2+m+6m+6}{6}\right)$$

$$= (m+1).\left(\frac{2m^2+7m+6}{6}\right) = \frac{(m+1).(m+2)(2m+3)}{6}$$

RHS $= \frac{(m+1)(m+2)(2m+2+1)}{6} =$ RHS

$= (m+1)($

6. $\therefore P(n)$ is true for $n = m+1$ whenever it is true for $n = 1$ and $n = m$.

Hence, $1^2 + 2^2 + 3^2 + \dots + n^2 = \dfrac{n(n+1)(2n+1)}{6}$

Prove by induction, $n^2 + n$ is even where n is positive integer.

Let, $P(n)$ be the statement where.

$P(n)$: $n^2 + n$ is even where n is positive integer.

Now, for $n = 1$

7. $1 + 1 = 2$ that is divisible by 2

So, it is an even no.

$\therefore P(n)$ is true for $n = 1$.

Let us assume that, $P(n)$ is true for $n = m$.

$m^2 + m = 2k$, where k is positive integer. — (1)

8. Now, for $n = m+1$,

$(m+1)^2 + m + 1 = m^2 + m + 2m + 1 + 1$

$= 2k + 2m + 2$

$= 2[k + m + 1]$

$= 2k_1$, where k_1 is positive integer which is divisible by 2.

So, even.

$\therefore P(n)$ is true for $n = m+1$ whenever it is true for $n = 1$ & $n = m$.

Hence, $n^2 + n$ is even where n is positive integer.

GCD Problem

Find two integers u and v satisfying

gcd (272, 119) = 272 u + 119 v

① gcd (272, 119)

a, b → integers.

a = bq + r , $0 \leq r < b$,

272 = 119 × 2 + 34

119 = 34 × 3 + [17] – last non zero remainder.

34 = 17 × 2 + 0

gcd (272, 119) = 17.

gcd (272, 119) = 17

= 119 − 34 × 3

= 119 − (272 − 119 × 2) × 3

= 119 − 272 × 3 + 119 × 6

= 119 (1 + 6) + 272 (−3)

= 119 · 7 + 272 · (−3)

= 119 v + 272 u

comparing we get. v = 7, u = −3

Find two integers u and v satisfying.

63 u + 55 v = 1

gcd (63, 55)

63 = 55 × 1 + 8

55 = 8 × 6 + 7

8 = 7 × 1 + [1] – last non. zero remainder

7 = 7 × 1 + 0

gcd (63, 55) = 1

6. $\gcd(63, 55) = 1$

$= 8 - 7 \times 1$

$= 8 - (55 - 8 \times 6)$

$= 8 - 55 + 8 \times 6$

$= 47 + (55 + 55 \times 1) \times 6$

$= 47 + 378 - 330$

$= 8 + 55 + (63 + 55 \times 1) \times 6$

$= 8 - 55 + 63 \times 6 - 55 \times 6$

$= 8 \times 7 - 55$

7. $= (63 - 55 \times 1) 7 - 55$

$= 63 \times 7 - 55 \times 7 - 55 \times 1$

$= 63 \times 7 + 55 \times (-8)$

$= 63u + 55v$

$u = 7, v = -8.$

8. # Find the $\gcd(595, 252)$ and express it in the form

$252m + 595n.$

$\gcd(595, 252) =$

7. $595 = 252 \times 2 + 91$

$252 = 91 \times 2 + 70$

$91 = 70 \times 1 + 21$

$70 = 21 \times 3 + \boxed{7}$ — last non-zero remainder

$21 = 7 \times 3 + 0$

$\gcd(595, 252) = 7.$

$252 \overline{)595} \; 2$, 504

$91 \overline{)252} \; 2$, 182

$70 \overline{)91} \; 1$, 70

$21 \overline{)70} \; 3$, 63

$7 \overline{)21} \; 3$, 21, 0

$$\gcd(595, 252) = 7$$

$$= 70 - (21 \times 3)$$

$$= 70 - (91 - 70 \times 1) \times 3$$

$$= 70 - 91 + 70 \times 3$$

$$= 70 \times 4 - 91$$

$$= (252 - 91 \times 2) 4 - 91$$

$$= 252 \times 4 - 91 \times 8 - 91$$

$$= 252 \times 4 - 91 \times 9$$

$$= 252 \times 4 - (595 - 252 \times 2) \times 9$$

$$= 252 \times 4 - 595 \times 9 + 252 \times 18$$

$$= 252 \times 22 + 595 \cdot (-9)$$

$$= 252m + 595n.$$

- If $\gcd(a, b) = 1$, then show that $\gcd(a^2 + b^2) = 1$

  * If gcd of two numbers of is 1, then they are common to each other.

Note: If two integers not both zero are prime to each other i.e. $\gcd(a, b) = 1$ if and only if, there exist 2 integers u & v such that

$$\boxed{au + bv = 1}$$

Since, $\gcd(a, b) = 1$ then there exist 2 integers, u & v such that. $au + bv = 1$

$$au = 1 - bv$$

$$a^2u^2 = (1 - bv)^2$$

$$a^2u^2 = 1 - 2bv + b^2v^2$$

$$a^2u^2 + 2bv - b^2v^2 = 1$$

$$a^2u^2 + b(2v - b^2u) = 1$$

$a^2p + bq = 1$ where $p = u^2$, $q = 2v - b^2u$ are some integers.

$bq = 1 - a^2p$

$b^2q^2 = (1-a^2p)^2$

$b^2q^2 = 1 - 2a^2p + a^4p^2$

$b^2q^2 + 2a^2p - a^4p^2 = 1$

$b^2n + a^2(2p - a^2p^2) = 1$

$b^2n + a^2m = 1$

where, $n = q^2$, $m = 2p - a^2p^2$ are some integers.

$\gcd(a^2, b^2) = 1$.

Congruence

Gauss introduced the concept of congruence.

Definition: Let m be a fixed positive integer. Two integers a and b are said to be congruent modulo m if $a-b$ is divisible by m. Symbolically expressed as $a \equiv b \pmod{m}$.

when $a-b$ is not divisible by m, a is said to be incongruent to b modulo m.

Properties:

1. $a \equiv a \pmod{m}$.
2. If $a \equiv b \pmod{m}$ then $b \equiv a \pmod{m}$.
3. If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$ then $a \equiv c \pmod{m}$.
4. If $a \equiv b \pmod{m}$ then, for any integer c;
   - $a+c \equiv b+c \pmod{m}$.
   - $ac \equiv bc \pmod{m}$.
5. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$
   - $a+c \equiv b+d \pmod{m}$.
   - $ac \equiv bd \pmod{m}$.
6. If $a \equiv b \pmod{m}$ then $a^n \equiv b^n \pmod{m}$ for all positive integers n.

   Note: the converse is not necessarily true.
7. If $ax \equiv ay \pmod{m}$ and a is prime to m then $x \equiv y \pmod{m}$.

● $a \equiv b \pmod m$ then b is said to be a residue of a modulo m.

$3 \pmod 9 = 0 \times 9 = 0$

$3 - 0 = 3$.

Find the least positive residue in $3^{36} \pmod{77}$

$3^4 \equiv 4 \pmod{77}$ — (i)

$3^{12} \equiv -13 \pmod{77}$ — (ii)

$(3^{12})^2 \equiv (-13)^2 \pmod{77}$

$3^{24} \equiv 15 \pmod{77}$ — (iii)

(ii) × (iii),

$3^{12} \cdot 3^{24} \equiv -13 \cdot 15 \pmod{77}$

$3^{36} \equiv -195 \pmod{77}$

$3^{36} \equiv -41 \pmod{77}$

$3^{36} \equiv 36 \pmod{77}$

$-41 + 36 = 77$

∴ The least positive remainder is 36.

Prove that. $19^{20} \equiv 1 \pmod{181}$

$19^2 \equiv -1 \pmod{181}$

$19^{20} \equiv (-1)^{10} \pmod{181}$

$\equiv 1 \pmod{181}$.

[Proved].

Use the congruence theory to prove that.

For all $n>1$ $17 \mid (2^{3n+1} + 3.5^{2n+1})$.

$= 2^{3n+1} + 3.5^{2n+1}$

$= 2^{3n} \cdot 2 + 3 \cdot 5^{2n} \cdot 5$

$= (2^3)^n \cdot 2 + 3 \cdot (5^2)^n \cdot 5$

$= 2.8^n + 15.25^n$

$25 \equiv 8 \pmod{17}$.

$25^n \equiv 8^n \pmod{17} \quad \forall n \geq 1$

$15.25^n \equiv 15.8^n \pmod{17}$

$15.25^n - 15.8^n \equiv 0$ — (i)

Now. $17.8^n \equiv 0 \pmod{17}$ — (ii)

(i) + (ii).

$15.25^n + 17.8^n \equiv 0 \pmod{17}$

$2.8^n + 15.25^n \equiv 0 \pmod{17}$

$2^{3n+1} + 3.5^{2n+1} \equiv 0 \pmod{17}$.

$\therefore 17 \mid (2^{3n+1} + 3.5^{2n+1}) \quad \forall n > 1$

Solution of linear Congruence :

Solve the linear congruence $5x \equiv 3 \pmod{11}$

The general form : $ax \equiv b \pmod{m}$

(i) $\gcd(a, m) = 1 \Rightarrow$ unique solution.

(ii) $\gcd(a, m) = d$ & d does not divide $b \Rightarrow$ no solution.

(iii) $\gcd(a, m) = d$ & d divides $b \Rightarrow$ many solution.

no. of solution $\rightarrow d \pmod{m}$.

$a = 5,\ b = 3,\ m = 11$

$\gcd(5, 11) = 1$

hence the linear congruence has a unique solution.

Since, $\gcd(5, 11) = 1$ then there exists two integers such that $5u + 11v = 1$

$5 \cdot (-2) + 11(+1) = 1$

$5 \cdot (-2) \equiv 1 \pmod{11}$.

$5 \cdot (-6) \equiv 3 \pmod{11}$.

$\therefore x \equiv -6 \pmod{11}$ is the solution of the linear congruence.

$\Rightarrow x \equiv 5 \pmod{11}$.

— This the final solution.

#. Solve the linear congruence $6x \equiv 3 \pmod 9$

$ax \equiv b \pmod m$

$a = 6,\ b = 3,\ m = 9$

$\gcd(6, 9) = 3$ divided 3

Hence, the given linear congruence has many solution.

∴ No. of solution : $3 \pmod m$

$= 3 \pmod 9$

$= 3$

∴ The given linear congruence has three incongruent solution.

$6x \equiv 3 \pmod 9$ —— (i)

$2x \equiv 1 \pmod 3$ — (ii)

$a = 2,\ b = 1,\ m = 3$

$\gcd(2, 3) = 1$

∴ linear congruence (ii) has unique solution.

Then there exist two integers u and v such that

$2u + 3v = 1$

here, $u = -1,\ v = 1$

$\therefore 2(-1) + 3 \cdot 1 = 1$

$2 \cdot (-1) \equiv 1 \pmod 3$.

∴ ~~There~~ $x = -1$ is an incongruent solution of congruence (ii).

$x_k = x_0 + k\left(\frac{m}{d}\right)$ $\qquad m/d = 9/3 = 3$

$x_1 = x_0 + 1 \cdot m/d$

$= -1 + 3 = 2$

$x_2 = -1 + 2 \cdot 3$

$= 5$

∴ All the three solution are

$x = -1, 2, 5$.